Chemical Kinetic Assignment

1. Starting with the rate law of a 0th order reaFor ction $A \rightarrow P$, derive the expression for the time-dependence of the reactant concentration, half-life. How much time does the completion of a 0th order reaction take?
2. Deduce the expression for t_{max} (induction period) for a 2-step consecutive unimolecular reaction.
3. A reaction $A \rightarrow P$ proceeds via a single elementary step. The reaction half life is doubled if the initial reactant concentration is halved. What is the molecularity of the reaction?
4. For a 2-step consecutive unimolecular reaction ($A \leftrightarrow B \leftrightarrow C$), both steps are reversible with rate constants k_1, k_{-1} and k_2, k_{-2} (k_1, k_2 are forward and k_{-1}, k_{-2} are backward rate constants for steps 1 and 2 respectively). Using steady-state approximation show that the overall rate law is reduced to a single step $A \leftrightarrow C$ with forward and reverse rate constants k_{eff} and k'_{eff}, which can be expressed in terms of k_1, k_{-1}, k_2 and k_{-2}